The Additive Manufacturing Supply Chain 251

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Class Outline

Supply Chains and Supply Chain Management
Additive Manufacturing in the Supply Chain
AM Supply Chain: Material Costs
AM Supply Chain: Competitiveness
AM Supply Chain: Responsiveness
Review: Supply Chain Management and Additive Manufacturing
Materials Shipping and Storage
Material Supplier Requirements
Review: AM Supply Chain

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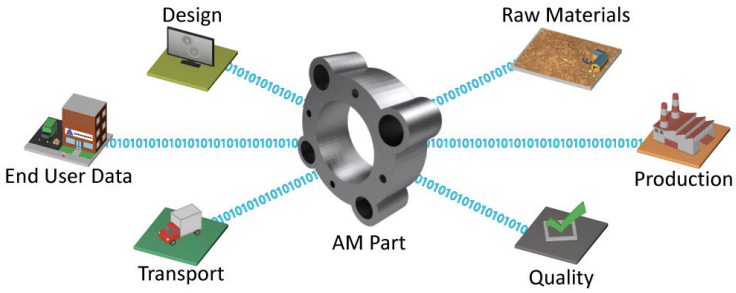
Supply Chains and Supply Chain Management



Every product that reaches a customer is the result of a complex series of exchanges between many different businesses or facilities. For example, one business might supply the raw materials necessary to make a product, and another might create parts from those raw materials. A different business or facility might assemble the parts into a finished product and send them to yet another business that stores and distributes the finished product. Finally, a different business might sell the product to customers. These companies that exchange information, goods, and services in order to create and sell a specific product form a supply chain. Typical supply chains within manufacturing often consist of raw material suppliers, product designers and engineers, manufacturing facilities, warehouses and distribution centers, shipping companies, and retailers, all of which are scattered throughout the world.

A manufacturer's success often depends on its supply chain providing a continuous flow of materials, products, and information. As a result, supply chain management has become a necessary part of manufacturing. Supply chain management involves coordinating a variety of activities, from initial material purchases to final product distribution.

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Additive Manufacturing in the Supply Chain

Additive manufacturing (AM) has the potential to revolutionize traditional supply chains and their management. AM technology turns digital three-dimensional (3D) models into physical objects by building them up in layers. In additive manufacturing, engineers create a part design and then convert it to a 3D computer model. An AM machine turns the computer model into a finished part by stacking and joining layers of material. End-use AM parts often have improved complexity and functionality compared to traditionally manufactured parts. AM may introduce several advantages to supply chain management, possibly including waste reduction, the combination of production and assembly into fewer steps, minimized storage needs, and increasing responsiveness to customers.

Though adopting AM in a facility may have a high up-front cost, supply chain managers must consider the affects AM may have on their supply chain as well as the total cost of ownership (TCO) of new programs and devices. TCO is the purchase price of an asset plus the costs of operation. If using AM processes allows a manufacturer to produce more products in less time, or to spend less on materials and storage, then the up-front cost will be balanced by reduced daily costs. By evaluating TCO, a supply chain manager can decide if additive manufacturing programs will prove to be efficient investments over time.

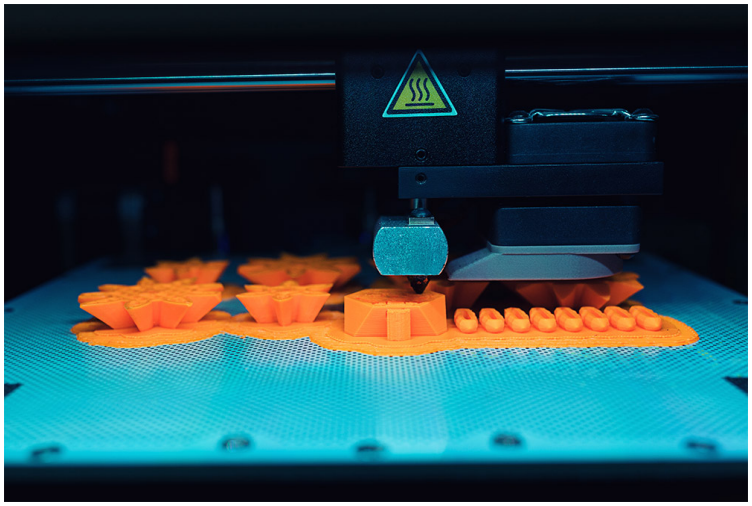

Figure 1:

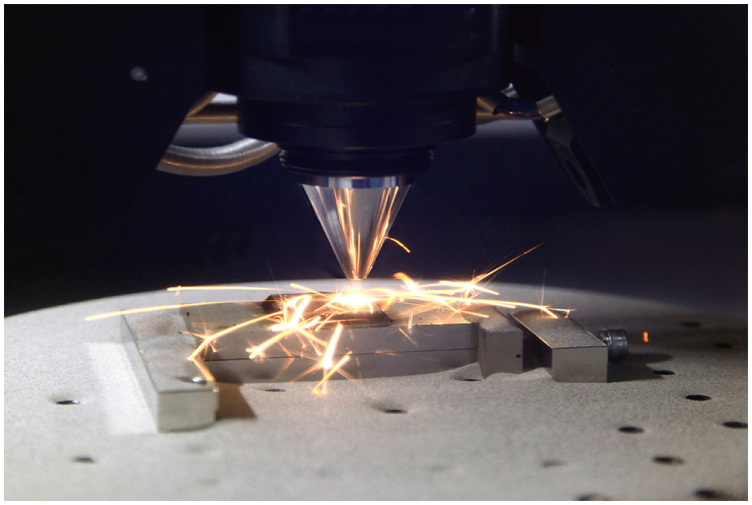

Figure 2:

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AM Supply Chain: Material Costs

Compared to traditional manufacturing, AM allows companies to reduce both the amount of materials required to create a part and the costs associated with shipping and receiving raw materials.

Traditional machining processes are subtractive. In other words, they create parts by removing small chips from pieces of material. Removed chips are generally waste, usually unable to be used in further processes. Additionally, most traditional manufacturing processes must use more material to create products with increased design complexity because part complexity is restricted by the shape of the raw material or a tool's ability to reach certain workpiece areas. As a result, creating parts with greater complexity can create increased waste and sometimes be prohibitively expensive.

By contrast, AM processes build parts by joining together layers of material, so there is little to no waste. Some AM parts require support structures, which are waste, though manufacturers can generally recycle any excess build material for use in another AM process.

AM technology can also generally create parts with complex features more easily and efficiently than traditional manufacturing techniques. For example, additive manufacturing may require less assembly, as certain components can be built up together layer by layer rather than joined together after the fact. With AM, a more complex design does not necessarily require more material.

Because of its potential to reduce waste without sacrificing complexity, additive manufacturing may be useful to companies wishing to adopt lean manufacturing. Lean manufacturing is a methodology that focuses on minimizing waste within manufacturing systems while also maximizing productivity.

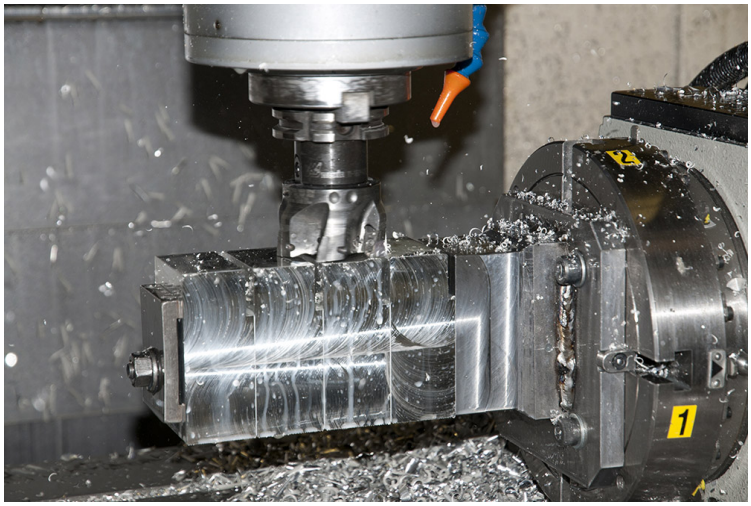

Figure 1: Traditional manufacturing is subtractive.

Figure 2: Material can be reclaimed and recycled in some additive manufacturing processes.

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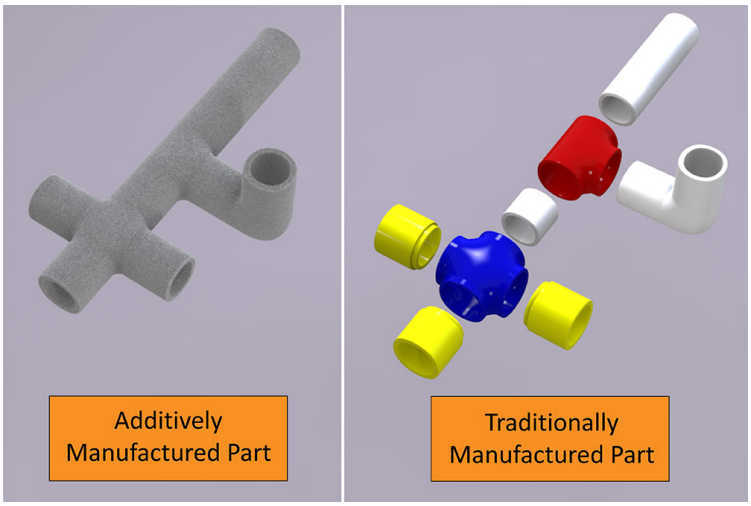
AM Supply Chain: Competitiveness

Supply chain management seeks to sustain competitive advantage. Additive manufacturing can provide this for those companies who can differentiate themselves through its use, even when used as a supplement to traditional manufacturing. Two common ways that AM can supplement traditional manufacturing are through customized manufacturing aids and reduced assembly needs.

AM Gripper

AM can produce complex parts such as manufacturing aids, fixtures, and tools. These devices can enhance, speed up, perfect, or fix a manufacturing or assembly process. A small AM part may also be added to a traditionally manufactured assembly, such as building an AM gripper for a robotic arm created through a metal cutting or metal forming operation. These combinations take advantage of AM's ability to create intricate features and traditional manufacturing's ability to rapidly shape large, simple workpieces. Half of all supply chain leaders are using AM to support other manufacturing processes.



Part Integration

The use of AM turns assembly into a single-piece system, eliminating assembly costs. Components can be built up together rather than joined together after the fact. This eliminates or greatly reduces the need for the assembly piece of the supply chain.

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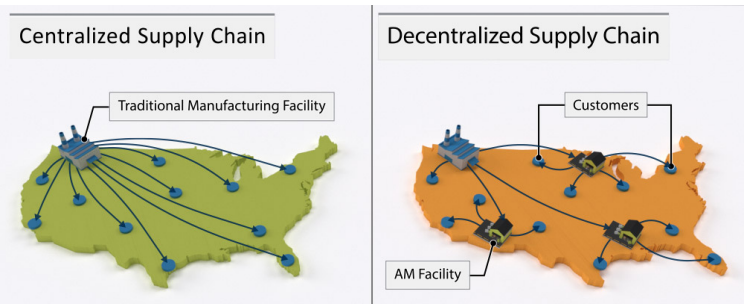
AM Supply Chain: Responsiveness



Since additive manufacturing increases manufacturers' ability to produce, assemble, and store parts within one facility, it reduces the need for centralization. As a result, AM processes can meet customers' needs more efficiently through distributed production in different locations. For example, rather than one centralized network of facilities shipping products to all customers, some of whom may be hundreds or thousands of miles away, a company can use multiple additive manufacturing facilities in different locations. Producing objects near customers reduces the need for part storage, decreases shipping costs, and greatly reduces lead time. Distributed production is flexible enough to respond to unpredictable customer demand without increasing costs for transportation and other logistics.

Another way that AM processes allow for greater responsiveness than traditional manufacturing is through quickly replenishing parts. A manufacturer need only change the model file, rather than the whole machine, to create new parts. Because of this, there is no longer a need for safety stock or warehouses to store that stock. While manufacturers will still need to store raw materials, doing so generally requires much less space than storing finished goods.

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Review: Supply Chain Management and Additive Manufacturing

1. The supply chain includes the source of raw materials. **TRUE**
2. Additive manufacturing machines a piece of material down to a part. **FALSE**
3. Total cost of ownership is the purchase price of an asset plus the costs of operation. **TRUE**
4. Lean manufacturing maximizes productivity without concern for waste. **FALSE**
5. On-demand production increases responsiveness to customers. **TRUE**
6. Additive manufacturing can reduce the need for safety stock. **TRUE**

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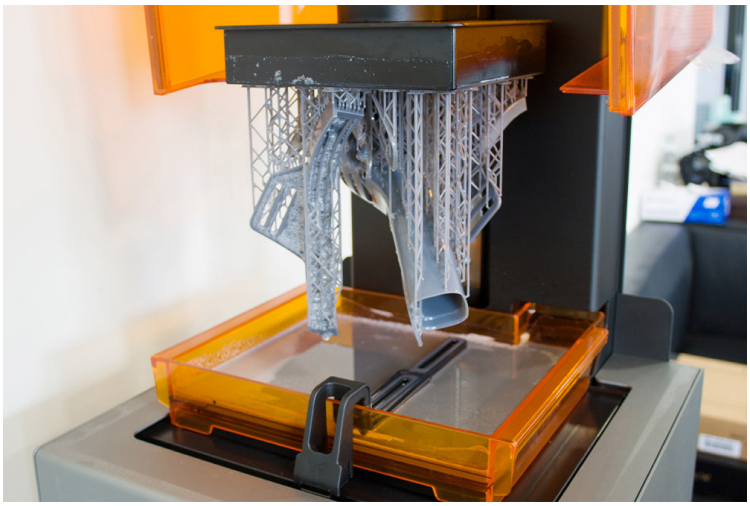
Materials Shipping and Storage

For example, some additive manufacturing relies on plastic polymers, such as nylon. When polymers are poorly stored, they encounter moisture from the air and later break down during AM processes, resulting in a weakened or defective finished product. Storage that reduces moisture, such as vacuum sealing, will prevent this.



Resin polymers are another common material used in AM. Resin refers to a solid or highly viscous substance of plant or synthetic origin. Due to its high light-sensitivity, resin can be rendered unusable for manufacturing by exposure to UV light. Resin must be kept tightly sealed in dark containers with a small headspace of air to prevent it becoming solid.

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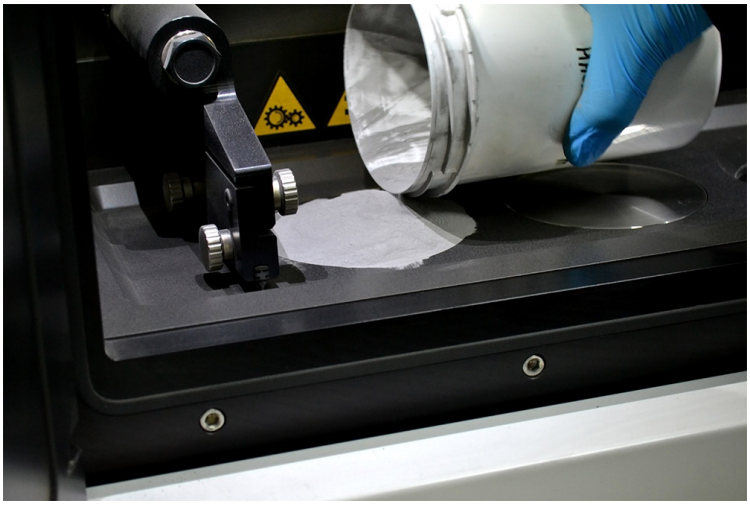

Stainless steel, a metal alloy, is used in additive manufacturing in the form of fine powder. If it becomes airborne, this powder represents a health hazard, either from inhaling or absorption through the skin. For many metal powders, fires and explosions are also a real concern for improperly handled materials. For these reasons, manufacturers must use special storage and containment procedures for metal powders, such as inert transfer. An inert transfer system, powered by vacuums, prevents the powders from coming into contact with oxygen.

Other materials used in AM include different plastics, titanium, gold, silver, and ceramics. Each material type has unique storage needs that supply chain managers must investigate and plan to adhere to before procuring the material.

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Material Supplier Requirements

Additive manufacturing processes can be time-consuming depending on the part size and design. Manufacturers can optimize AM processes by selecting the best materials for production. Doing so is a crucial earlier step in supply chain management.

Material selection is important not only for part performance, but to make sure production processes and end-use parts are in compliance with applicable laws and regulations. Historically, additive manufacturing was primarily used in prototyping, or creating parts to test their design or function. Because prototypes are not used as finished products, early adopters of AM selected materials without much concern for regulations. Today, however, additive manufacturers must ensure regulatory compliance. Parts for industries like aerospace, automotive, and medical devices may pose serious safety hazards if they are made incorrectly. A faulty automotive component, for example, can lead to a fatal traffic accident. Regulations for additive manufacturing ensure that parts are made safely.

Material supply compliance can be tracked throughout the supply chain using the digital thread. The digital thread is the framework that maintains data on a part or machine to produce a comprehensive view of its movement throughout its lifecycle. This framework addresses protocols, security, and standards. Digital threads often work in conjunction with a digital twin, which is a virtual representation of a physical object.

Figure 1: Additive manufacturing of medical implants requires several levels of regulatory compliance.

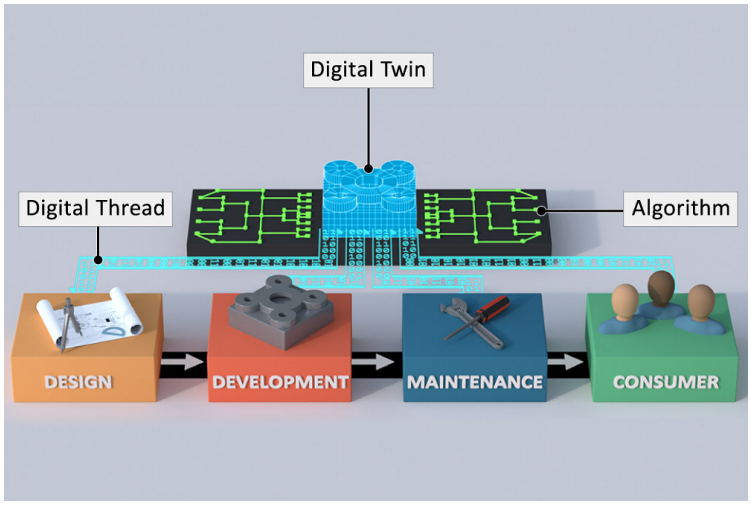

Figure 2: The digital thread can track elements throughout the supply chain.

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Review: AM Supply Chain

- **Digital thread**: A framework that tracks an element throughout its lifecycle in the supply chain.
- **Manufacturing aid**: Any device, platform, tool, or fixture used to enhance or fix manufacturing or assembly production.
- **Regulatory compliance**: When a business follows state, federal, and international laws and regulations for its operations.
- **Safety stock**: Extra materials or products that are maintained to mitigate risks of uncertain supply and demand.

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Class Vocabulary

Additive Manufacturing AM. The process of joining or solidifying materials to make an object based on a three-dimensional computer model. Additive manufacturing methods typically build up layers of material to create an object.

AM Additive manufacturing. The process of joining or solidifying materials to make an object based on a three-dimensional computer model. AM methods typically build up layers of material to create an object.

Assembly The action of fitting together the component parts of a machine or other object. Assembly is simplified by and requires less labor with additive manufacturing.

Decentralization The movement of departments of a large organization away from a single administrative center to other locations. Decentralization allows for elements of the supply chain to be spread out in different locations.

Digital Thread An integrated view of all the data and information about a part or product throughout its lifecycle. The digital thread connects information from all aspects of a product into one seamless network.

Digital Twin A virtual representation of a physical object, such as a part or machine. A digital twin evolves with the object throughout its lifecycle.

Distributed Production Manufacturing that occurs in separate locations rather than at one centralized facility. Distributed production can occur closer to the end users and thus be more responsive to their needs.

End-Use Designed to be used directly by a consumer or directly in another manufactured product. End-use products created by additive manufacturing (AM) include medical implants, custom dental devices, and camera equipment.

Fixtures A piece of equipment that is fixed into position. Fixtures can be efficiently produced by additive manufacturing.

Gripper A mechanical device that uses rotary and linear actuators to mimic the motion of the human hand. A gripper has a complex shape that can be easily produced using additive manufacturing.

Inert Transfer A containment procedure powered by vacuums that prevents powders from coming into contact with oxygen. Inert transfer is used with metal powders for additive manufacturing.

Lead Time The time between the initiation and completion of a production process. Lead time is shortened in additive manufacturing.

Lean Manufacturing A methodology that focuses on minimizing waste within manufacturing systems while also maximizing productivity. Lean manufacturing exploits all opportunities to safely eliminate waste.

Lifecycle The entire timeline of something. In manufacturing, lifecycle includes part design, machine setup, production, quality, and end of life.

Manufacturing Aids Any device, platform, tool, or fixture used to enhance, speed up, perfect, or fix a manufacturing or assembly process. Manufacturing aids can be efficiently produced by additive manufacturing.

Nylon A synthetic plastic, the most common plastic material used in AM. Nylon's advantages include its durability and excellent strength-to-flexibility ratio.

Prototyping Creating an early sample, model, or release of a product built to test a concept or process. Early additive manufacturing was primarily used for prototyping.

Regulatory Compliance When a business follows state, federal, and international laws and regulations relevant to its operations. Regulatory compliance varies by industry, from aerospace to medical devices to many other fields.

Resin A solid or highly viscous substance of plant or synthetic origin that is typically converted into polymers. Resin's advantages include its low shrinkage rate and its high chemical resistance.

Responsiveness The quality of reacting quickly and positively. Additive manufacturing can increase the responsiveness of the supply chain.

Robotic Arm A programmable or remote-controlled mechanical device that simulates the movement of a human arm. Robotic arms are used in a variety of assembly and manufacturing applications.

Safety Stock Extra materials or products that are maintained to mitigate risks of uncertain supply and demand. The need for safety stock is decreased or eliminated by additive manufacturing.

Stainless Steel A metal alloy used in fusion or laser sintering. Stainless steel used in additive manufacturing usually takes the form of a powder blend.

Subtractive A type of manufacturing that involves a piece of material being machined down to a part. Traditional manufacturing is subtractive.

Supply Chain A network of companies that exchange resources such as materials and information to deliver products to customers. A supply chain consists of a company, its suppliers, its distributors, and its customers.

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Supply Chain Management The process of planning, implementing, and controlling supply chain activities to achieve maximum customer value and sustain competitive advantage. Supply chain management oversees each organization in the supply chain, from development to sourcing to production to delivery.

Support Structures A reinforcing component used to hold the weight of an additively manufactured part as it is being constructed. Support structures are removed from the part once the build is complete.

Three-Dimensional 3D. Having height, width, and depth. Three-dimensional parts are created during additive manufacturing (AM) processes.

Total Cost Of Ownership TCO. The purchase price of an asset plus the costs of operation. Total cost of ownership helps assess the value of an investment over time.

Virtual Existing on or simulated within a computer program or system rather than physically existing. Virtual representations of real-world objects are known as digital twins.

Waste Unused material or the byproducts of production. Waste is to be avoided in lean manufacturing.

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